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It is said that science fiction inspires reality. In fact, the pace of technological transformation over the last two decades has been astonishing. It reached a crescendo, catapulting us to a whole new level of connectivity, communication, and data management. Who knows what this confluence of data and technology will deliver to us by the end of this decade?

As the world finds its feet, operating in the digital and virtual space, we are seeing machines becoming more intelligent, bots reacting in more human ways, and connectivity become more ubiquitous. All of this is fueling the generation of new data that has the potential to give us a whole new perspective, if analyzed, understood, and applied correctly. In fact, according to the IDC Global DataSphere forecast, over the next three years, the world is expected to generate more data than it did in the past three decades.

When combined with the latest developments in technology, this data explosion across the world is set to change the game radically. Let us take a quick look at some key areas to see how this is likely to happen:

In the age of Edge Computing and 5G

5G is all about connecting things reliably, without lag, in real time. Take for example autonomous cars. The complexity of making real-time deci-

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Data and Technology Trends that will Shape our Future

sions was stalling the fully autonomous vehicle. With 5G, it will become possible to track, collate, analyze, and apply data just as a human driver would in a variety of driving conditions. As technology integrates more deeply with the systems and processes, it is natural to rely on Edge Computing that brings computation and data storage closer to the end-destination to improve response times.

What's more, the industry is looking at using virtual fleets of autonomous vehicles to generate relevant data for real-time trials. A fleet of 100 simulated vehicles could generate as much as 200 to 500 petabytes of raw data. This massive amount of data will necessitate a rethink on the part of the manufacturers regarding the infrastructure they will require for managing this volume of data that needs to be durable, and at the same time be easy to retrieve. 5G, and taking the thought logically further even 7G in the future, could make handling, storing, and retrieving this magnitude of data easy.

Intelligent Interfaces Powered by Artificial Intelligence (AI) and Machine Learning (ML)

Today, it is fairly common to connect to a chatbot when making a service-related call. From banking to hospitality to even healthcare, intelligent interfaces make it easy for you to not only get the information that you want, but also assist you in taking relevant decisions related to it. Taking this thought further into the future, these intelligent interfaces are set to become a ubiquitous part of our lives.

Take the case of the healthcare industry, as the world is reeling from the impact of the pandemic, the healthcare providers are (in parallel) having to upgrade their legacy systems to pro-

vide services to not only those afflicted by the pandemic, but also those who require long-term care, specialty care, or just a consultation.

AI is also helping accelerate research and innovation in healthcare. For instance, there's work done to train an AI model to find cancerous difference and to ignore healthy differences, which is helping medical professionals help vulnerable groups.

As the world leverages virtual interfaces for ensuring business continuity, it is but logical to see the same transition to a smart factory. Adoption of fourth industrial revolution technologies to perform real-time analytics, using machine-to-machine connectivity to provide deeper insights has made the production line efficient.

What is this likely to mean for you a few years down the line? Imagine waking up to your wearable smart device giving you a quick update on your day's schedule and checking with you for your breakfast preferences, followed by a quick reminder about your upcoming virtual consultancy with your healthcare provider and a reminder to share your latest health records with them. All of this happens through verbal instructions without the need to 'login' to any device or enter a multi-digit password. This not only entails the generation of immense amounts of data, but also appropriate storage, management, and retrieval in order for the world to truly benefit.

The potential that data and technology hold to impact our world is only limited by our imagination. How far each of these and other data trends disrupt the world remains to be seen. However, individually, and collectively they will make a significant impact on the way we live and work, especially over the next decade or so. 